Chaokang Jiang Homepage · Scholar

+86 177-9881-7596 jck98@foxmail.com 1998.01 World-Model Algorithm

Education

2020.09 ∼ 2023.06 M.Sc.·Control Sci.&Eng.·SJTU joint program Advisors: H. Wang (Dean, Pujiang Int'l), Y. Miao IRMV Lab

Work Experience

2023.06 ∼ 2025.03	Perception Algorithm	Phigent Robotics	Beijing
2025.03 ∼ *Present*	World-Model Algorithm (top group perf.)	Bosch (XC-CN)	Shanghai

Project Experience

- **Generative Closed-Loop AD Simulation** — Bosch (XC-CN) · 2026.01 ∼ *Present*

  1) Designed a 7V surround video generator over the full Map→WorldSim→7V pipeline, distilled for **13.9**× faster inference; **2)** OmniDream delivered the **industry-first**, near-real-time 7V streaming generation; **3)** built a Smart-Agent vector world model for controllable closed-loop sim.

- **Two-/One-Stage Perception Iteration** — Bosch (XC-CN) · 2025.03 ∼ 2025.12

  1) Built and iterated a static-perception OneModel (3→7 features), owning its data loop and releases; **2)** delivered a DYO & Static two-in-one OneModel (one-stage); **3)** led dynamic-model quantization and Mono3D releases.

- **Controllable Surround Video Generation** — Phigent Robotics · 2024.07 ∼ 2025.03

  1) Single-frame: extended MagicDrive to internal rigs (4 fisheye/7V/11V) with 3D-BBox/map labels for data gen and editing; **2)** Video: on OpenSora-STDiT + MagicDrive, built text-/layout-controllable surround video across views/res/length, up to **242+ frames**, validated on cone-lines and truck swaps.

- **Perception/Data-Gen Optimization & Sparse E2E PoC** — Phigent Robotics · 2024.02 ∼ 2024.08

  1) Iterated a BEVFusion dynamic auto-labeler (better VRU precision, long-range recall); **2)** optimized calibration QC and pure-LiDAR 3D-detection efficiency; **3)** built a Pillar 3D velocity-flow model on J6E/Orin with scene-flow labeling and point-cloud prediction; **4)** owned the 11V+LiDAR Sparse BEV-Fusion and fused CUDA ops.

- **Stereo Road-Preview Magic-Carpet (SOP)** — Phigent Robotics · 2023.06 ∼ 2024.04

  1) Deployed road-segmentation models on TDA4, Horizon J5/J6E and NVIDIA Orin (FP + QAT), balancing accuracy/speed into mass production; grew data 5W→**23W+** over 20+ releases; **2)** prototyped a stereo depth model in parallel.

- **More projects (see homepage, some encrypted: 5896)**

  8V pure-vision E2E, vector + sensor-level closed-loop sim, perception-planning-decision (aerospace), Positec lawn-mower.

Academic Achievement

- **10+ papers** across ICML (Spotlight), NeurIPS, CVPR, ICCV, AAAI and T-PAMI, TII, TIM, TITS, AIS (full list on Scholar).
- **VectorWorld** (**ICML Spotlight**) Streaming vector world model for AD, dual SOTA in accuracy and efficiency; cited in a keynote by the President of Bosch Intelligent Driving Control China.
- **3DSFLabelling** (**CVPR 2024**) & **NeuroGauss4D-PCI** (**NeurIPS 2024**) ultra-precise 3D motion-flow auto-labeling and 4D point-cloud interpolation/prediction; both SOTA.
- **Pseudo-LiDAR Scene Flow** (**TII**, Q1) First to estimate 3D scene flow from monocular vision via a depth-consistency loss.
- **More:** DiffFlow3D(CVPR'24/T-PAMI), Mamba4D(CVPR'25), RegFormer(ICCV'23), TransLO(AAAI'23), Pseudo-LiDAR VO(TIM'23), SFGAN(AIS'22) — scene flow, LiDAR odometry/registration, 3D/4D perception.

Expertise

PyTorch, mmdet, diffusers, transformers, CUDA, Docker, PCL; Vibe-Coding; R&D-to-production; 10+ NSFC (general/youth) grants.